

ObesitySense

Obesity prediction and Analysis using Machine Learning

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2 INTRODUCTION

2.1 BACKGROUND

Obesity is one of the most significant global health issues of the 21st century. According to the World Health Organization (WHO, 2021), worldwide obesity rates have nearly tripled since 1975, contributing to the rise in heart diseases, diabetes, and various forms of cancer. Traditional methods of assessing obesity risk mainly rely on body mass index (BMI). However, with the increasing availability of health datasets and advancements in machine learning, it is now possible to predict obesity levels more accurately based on multiple lifestyle factors.

2.2 AIM

The aim of this project is to develop a machine learning based system that can predict an individual's obesity level based on their personal habits and physical attributes. This system is intended to help individuals and healthcare practitioners identify obesity risks early and recommend preventive actions.

2.3 OBJECTIVE

The main objectives of the project are:

1. To perform data preprocessing to prepare the dataset for machine learning models.
2. To explore the data thorough visualizations and correlation analysis to understand important factors contributing to obesity.
3. To train and evaluate machine learning models.
4. To deploy the best-performing model in an interactive Streamlit web application.

3 LITERATURE REVIEW

There are several studies in which machine learning models are used to predict obesity based on various factors like eating habits, physical activity, and lifestyle choices. For example, López et al. (2018) demonstrated that decision trees could be used effectively to classify individuals based on their obesity levels. Their model achieved an accuracy of 85%, which highlighted the potential of ML in this field.

Similarly, González et al. (2019) explored different classification techniques, improving prediction accuracy by combining feature selection with oversampling methods (like SMOTE). Their model achieved an impressive accuracy of 91%, showing that refining data preprocessing methods can significantly enhance the reliability of predictions.

Feature engineering and selection also play a crucial role in improving model performance. Zhang (2017) emphasizes that careful selection of relevant features and reduction of unnecessary complexity helps achieve better prediction accuracy.

4 DATASET

This dataset focuses on estimating obesity levels in individuals from Mexico, Peru, and Colombia. It includes 17 attributes and 2,111 records, with the target variable being NObeyesdad (Obesity Level). The obesity levels are classified into seven categories: Insufficient Weight, Normal Weight, Overweight Level I, Overweight Level II, Obesity Type I, Obesity Type II, and Obesity Type III.

The dataset is a mix of synthetic and real-world data, with 77% generated using the Weka tool and SMOTE filter, and 23% collected directly from users via a web platform. There are no missing values in the dataset but the features labelling/values are inconsistent.

This dataset was sourced from the UCI Machine Learning Repository and can be accessed via the following [link](#).

Table 1: Dataset column description

Feature	Feature Information	Possible Values/Category
Gender	Gender	Female Male
Age	Age	Numeric Value
Height	Height (in Meters)	Numeric Value
Weight	Weight (in KG)	Numeric Value
FAVC	Eat high caloric food frequently	YES NO
FCVC	Vegetables in meals	1-Never 2-Sometimes 3-Always
NCP	No. of main meals daily	1-Between 1 & 2 2-Three 3-More than three 4-No answer
CH2O	Water intake daily	1-Less than a litre 2-Btw 1 & 2 litre 3-More than 2 litre
CAEC	Food between meals	No Sometimes Frequently Always
family_history_with_overweight	Family history with overweight	No Yes
FAF	Physical activity frequency in a week	0-Never 1-One to two times 2-Two to four times 3-Four or five times
TUE	Technological device usage daily	0- 0 to 2 hours 1- 3 to 5 hours 2- More than 5 hours
SMOKE	Smoking	Yes No

SCC	Calorie monitoring	Yes No
CALC	Alcohol consumption	No Sometimes Frequently Always
MTRANS	Mode of Transportations	Automobile Motorbike Bike Public Transport Walking
NObeyesdad	Target classes (obesity levels)	Insufficient Weight Normal Weight Overweight level 1 Overweight level 2 Obesity type 1 Obesity type 2 Obesity type 3

5 EDA AND DATA PREPROCESSING

Throughout this project, I explored the data and applied preprocessing techniques as needed. I fixed issues as they appeared rather than doing EDA and preprocessing as two separate steps. This helped to keep things organised and well understood.

5.1 DESCRIPTIVE ANALYSIS

I performed some basic descriptive analysis to understand the dataset's structure, column names, data types and check missing values.

```

RangeIndex: 2111 entries, 0 to 2110
Data columns (total 17 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Gender                                2111 non-null   object
1   Age                                   2111 non-null   float64
2   Height                               2111 non-null   float64
3   Weight                               2111 non-null   float64
4   family_history_with_overweight       2111 non-null   object
5   FAVC                                  2111 non-null   object
6   FCVC                                  2111 non-null   float64
7   NCP                                   2111 non-null   float64
8   CAEC                                  2111 non-null   object
9   SMOKE                                 2111 non-null   object
10  CH2O                                  2111 non-null   float64
11  SCC                                   2111 non-null   object
12  FAF                                   2111 non-null   float64
13  TUE                                   2111 non-null   float64
14  CALC                                  2111 non-null   object
15  MTRANS                                2111 non-null   object
16  NObeyesdad                           2111 non-null   object

```

Figure 1: Columns info using `df.info()`

	Age	Height	Weight	FCVC	NCP	CH2O	FAF	TUE
count	2111.000000	2111.000000	2111.000000	2111.000000	2111.000000	2111.000000	2111.000000	2111.000000
mean	24.312600	1.701677	86.586058	2.419043	2.685628	2.008011	1.010298	0.657866
std	6.345968	0.093305	26.191172	0.533927	0.778039	0.612953	0.850592	0.608927
min	14.000000	1.450000	39.000000	1.000000	1.000000	1.000000	0.000000	0.000000
25%	19.947192	1.630000	65.473343	2.000000	2.658738	1.584812	0.124505	0.000000
50%	22.777890	1.700499	83.000000	2.385502	3.000000	2.000000	1.000000	0.625350
75%	26.000000	1.768464	107.430682	3.000000	3.000000	2.477420	1.666678	1.000000
max	61.000000	1.980000	173.000000	3.000000	4.000000	3.000000	3.000000	2.000000

Figure 2: Descriptive statistics using `df.describe()`

5.2 ASSIGNING CONSISTENT VALUES

There are two main issues with the values for some columns.

1. In some columns where the values should be categorical (such as never, sometimes always), there the values are numeric (such as 1,2,3) in columns like NCP, CH2O, FCVC etc.
2. And those values are not actually 1,2,3, these are like 1.00025, 2.00025, 3.000045 etc which could be due to some preprocessing or techniques applied while generating the synthetic data.

```
#checking unique values in for FCVC
print(df['FCVC'].describe())
print(df['FCVC'].value_counts().sort_index().head(20))
```

```
count    2111.000000
mean      2.419043
std       0.533927
min       1.000000
25%       2.000000
50%       2.385502
75%       3.000000
max       3.000000
Name: FCVC, dtype: float64
FCVC
1.000000    33
1.003566     1
1.005578     1
1.008760     1
1.031149     1
1.036159     1
1.036414     1
```

Figure 3: values in FCVC column

To fix this problem I rounded these values to the whole number and mapped correct categories as described in the dataset information from the [source](#).

```

#most of the columns which should have values like 1,2,3, have values like 1.0023,2.0045, so rounding them to the whole numbers
df['FCVC'] = df['FCVC'].round().astype(int)
df['NCP'] = df['NCP'].round().astype(int)
df['CH20'] = df['CH20'].round().astype(int)
df['FAF'] = df['FAF'].round().astype(int)
df['TUE'] = df['TUE'].round().astype(int)

#mapping the correct categories for features

fcvc_map = {1: 'Never', 2: 'Sometimes', 3: 'Always'}
df['FCVC'] = df['FCVC'].map(fcvc_map)

ncp_map = {1: 'Btwn 1 & 2', 2: "3", 3: 'More than 3', 4: "No Answer"}
df['NCP'] = df['NCP'].map(ncp_map)

ch2o_map = {1: 'less than 1L', 2: 'Btwn 1L & 2L', 3: 'More than 2L'}
df['CH20'] = df['CH20'].map(ch2o_map)

faf_map = {0: 'Never', 1: '1 to 2times', 2: '2 to 4 times', 3: "4 or 5 times"}
df['FAF'] = df['FAF'].map(faf_map)

tue_map = {0: '0 to 2h', 1: '3 to 5h', 2: 'More than 5h'}
df['TUE'] = df['TUE'].map(tue_map)

```

Figure 4: rounding the values and assigning correct categories.

5.3 CLASS IMBALANCE

To make sure my model does not get trained to be biased data towards certain classes, I checked if all the 7 classes have similar number of samples present. I used matplotlib and seaborn to create a bar chart.

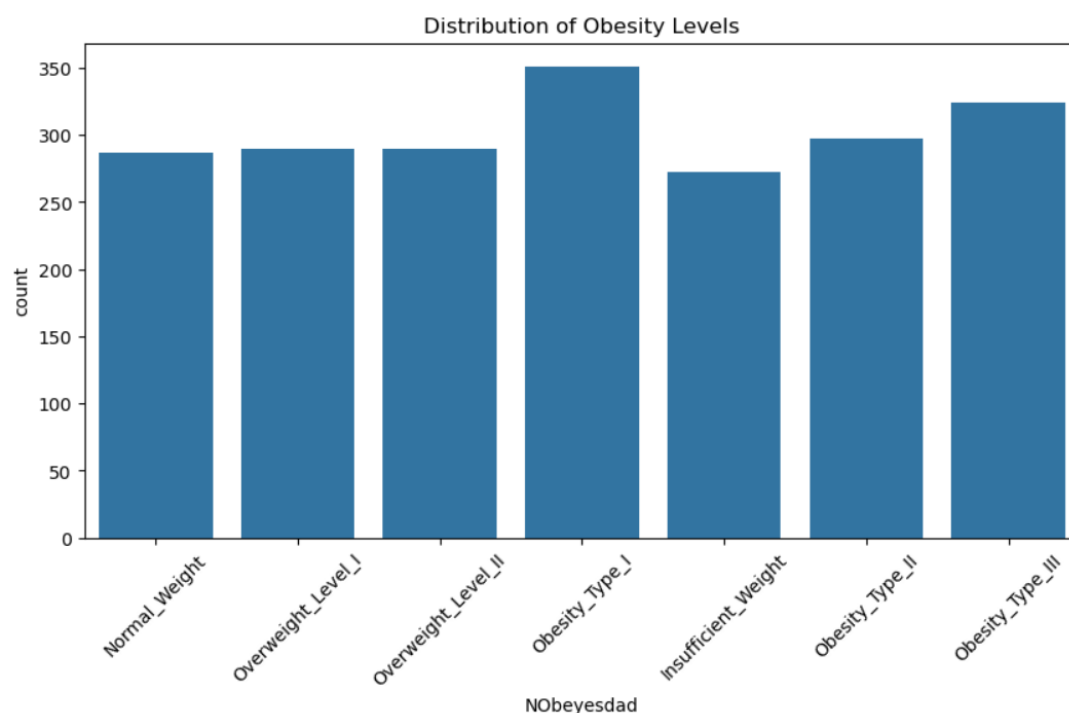


Figure 5: Samples per class

It is evident that all the classes are almost balanced, and we do not need to do any oversampling here. But to go deeper, I checked the class imbalance with respect to gender.

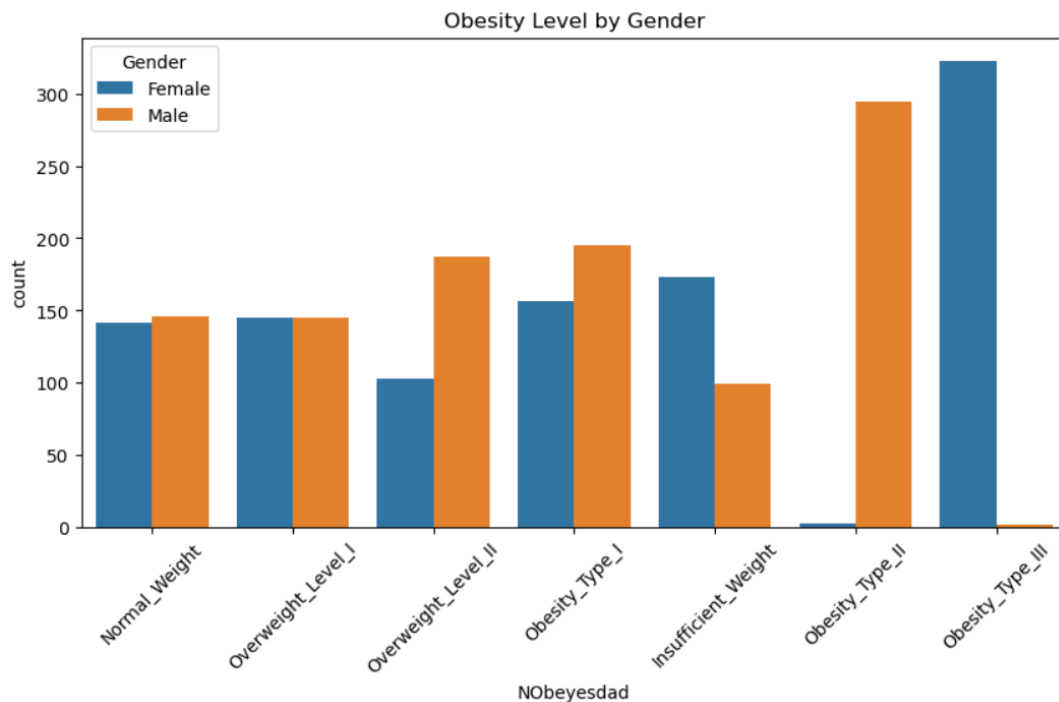


Figure 6: Samples per gender per class.

As it can be clearly seen, obesity type 2 and type 3 are imbalanced for male and female.

5.3.1 SMOTE

To balance the imbalanced class with respect to gender I tried using smote but SMOTE balances the classes for overall dataset (which is already balanced) not for gender specifically.

Therefore, I split the dataset into two dataframes, one of females and other for males. Then I applied SMOTE on both dataframes.

```
#balancing imbalanced class with respect to gender using SMOTE

from imblearn.over_sampling import SMOTE
#creating separate dataframes for males and females
df_male = df_scaled_encoded[df_scaled_encoded["Gender"]==1]
df_female = df_scaled_encoded[df_scaled_encoded["Gender"]==0]

#seperating features and target column
x_male = df_male.drop(columns = "NObeyesdad")
y_male = df_male["NObeyesdad"]

print("Male Class Distribution:")
print(y_male.value_counts())

#resampling
smote = SMOTE(random_state = 42, k_neighbors = 1)
x_male_resampled, y_male_resampled = smote.fit_resample(x_male, y_male)

#seperating features and target column
x_female = df_female.drop(columns = "NObeyesdad")
y_female = df_female["NObeyesdad"]

print("Female Class Distribution:")
print(y_female.value_counts())
x_female_resampled, y_female_resampled = smote.fit_resample(x_female, y_female)
```

Figure 7: Splitting the data and Applying SMOTE

But there was a ValueError because SMOTE expects atleast 2 samples for the minority class to oversample it but there was only 1 sample present for class 4.

Male Class Distribution:	
NObeyesdad	
3	295
2	195
6	187
1	146
5	145
0	99
4	1

Figure 8: Male Class distribution

5.4 ENCODING AND SCALING

- **Label Encoder:** For the target variable (NObeyesdad) I used Label encoder.
- **OneHotEncoder:** As MTRANS is a nominal feature, I used OneHotEncoding for that to make sure the model does not learn any order for this feature.
- **OrdinalEncoder:** For rest of the categorical ordinal and binary (yes/no) features I used OrdinalEncoder.
- **StandardScaler:** I scaled the continuous features such as Age, Weight and Height using StandardScaler.

All the encoders and scalers were saved using joblib for further use.

5.5 SKEWNESS

While exploring the dataset, I checked the distribution of continuous features (Age, Height, and Weight) and noticed that Age was highly right skewed(positive skewness) with a skewness of around 1.5, which could affect the performance of my model.

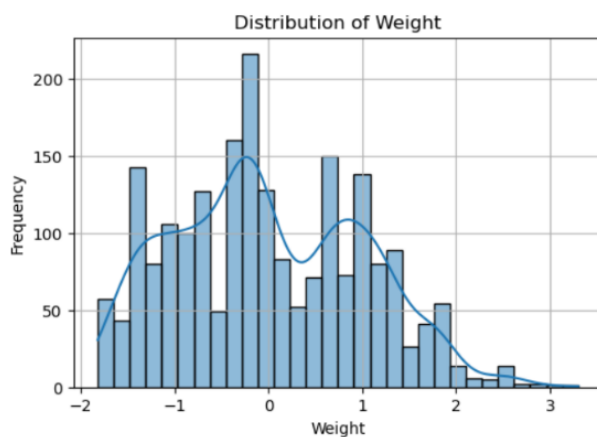


Figure 9: Weight Distribution

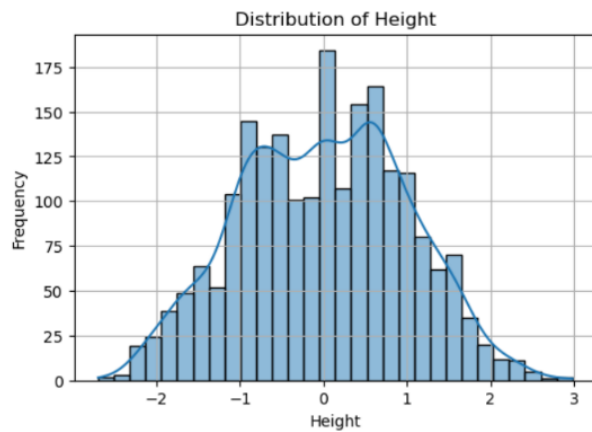


Figure 10: Height Distribution

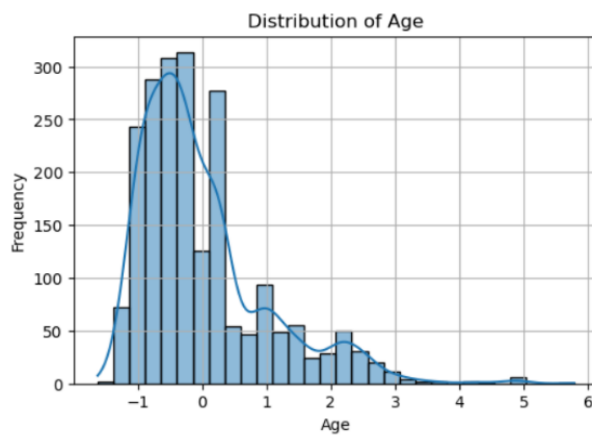


Figure 11: Age Distribution

5.5.1 Yeo-johnson Transformer

To handle the skewness in Age column I applied yeo-johnson transformation using powertransformer and reduced the skewness to 0.1.

```
#to check skewness again
df_scaled_encoded[['Age', 'Height', 'Weight']].skew()

Age      0.105573
Height   -0.012855
Weight    0.255410
dtype: float64
```

Figure 12: Skewness after transformation.

5.6 OUTLIER DETECTION AND HANDLING

To identify outliers in Age, Height, and Weight, I first created box plots. Then, I used the Interquartile Range (IQR) method to handle them. In this method, any value lying below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ was considered an outlier and removed.

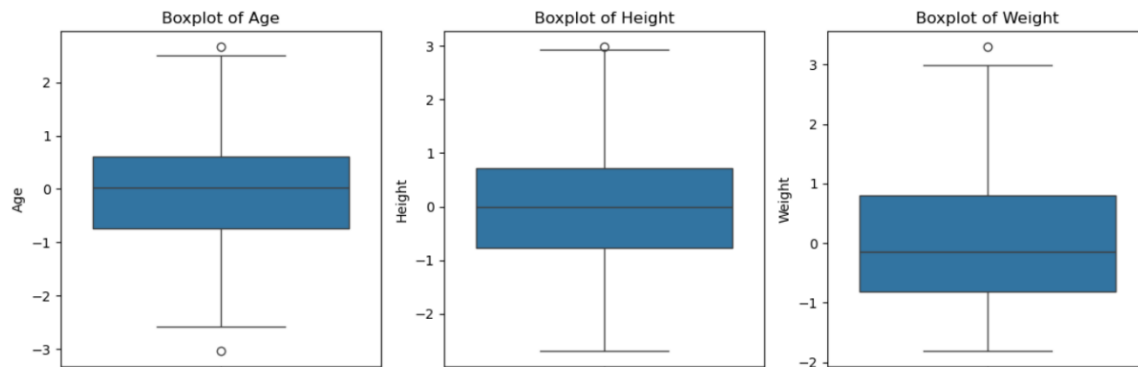


Figure 13: Boxplots for Age, Height and Weight.

5.7 CORRELATION ANALYSIS

To understand the relationship between the features I generated a correlation matrix and visualised it as a heatmap for better understanding. Along with the continuous features I also included encoded ordinal features in the hope of gaining some more information.

From the heatmap I noticed that Weight and Height have the strongest positive link (0.46), meaning taller people often weigh more which is obvious. Weight also has a medium link with NObeyesdad (0.39) and FAVC (0.27), showing connections eating habits. Age and NObeyesdad (0.28) suggest older people might have higher NObeyesdad values. On the other hand, CAEC has a medium negative link with NObeyesdad (-0.33) and Weight (-0.29).

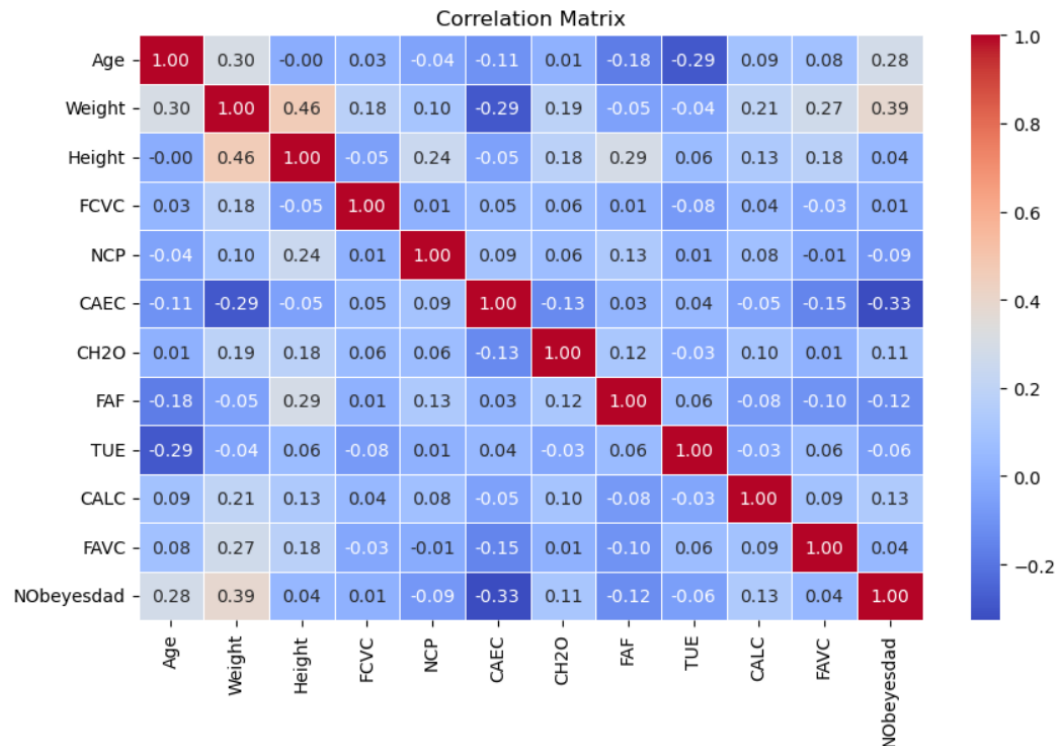


Figure 14: Correlation Heatmap

5.8 PRINCIPAL COMPONENT ANALYSIS (PCA)

I applied Principal Component Analysis (PCA) to reduce the dimensionality of the dataset to 2 principal components for visualization purposes. This helped me understanding how different obesity categories are distributed in a two dimensional space.

However, I used the PCA transformed data only for visual analysis and not for model training, as the original features had more meaningful information for prediction purposes.

From the scatter plot it is evident that most data point mix together in the middle, so it's hard to tell the obesity levels apart. But Normal Weight (green) and Obesity Type III (light purple) are a little more spread out on the sides, showing they might be more different from the rest. Overall, this means PCA might not fully separate these groups with just two components.

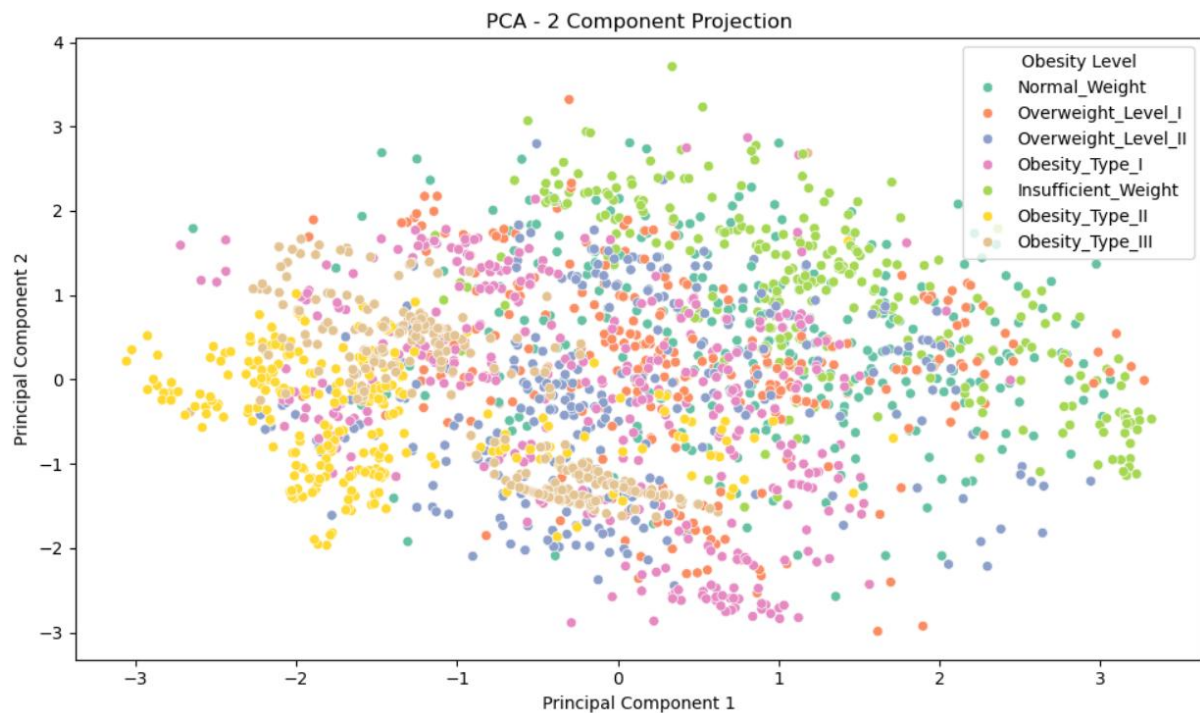


Figure 15: PCA transformed data distribution.

6 MODEL TRAINING AND EVALUATION

6.1 TRAIN TEST SPLIT

I split the data into 80% for training and 20% for testing and also kept stratified sampling on for target classes to have unbiased splits.

6.2 TRAINING RANDOMFOREST CLASSIFIER

I trained the RandomForest classifier with `n_estimators = 100`, `random_state = 42`, `class_weight = "balanced"`.

6.3 MODEL EVALUATION

Without using any optimisation technique my model achieved accuracy of 94% as tested on the test set. Also, the average precision, recall and f1 score was 94%.

	precision	recall	f1-score	support
Insufficient_Weight	0.98	0.95	0.96	55
Normal_Weight	0.78	0.89	0.84	57
Obesity_Type_I	0.96	0.96	0.96	70
Obesity_Type_II	1.00	0.98	0.99	59
Obesity_Type_III	1.00	1.00	1.00	65
Overweight_Level_I	0.89	0.88	0.89	58
Overweight_Level_II	0.98	0.91	0.95	58
accuracy			0.94	422
macro avg	0.94	0.94	0.94	422
weighted avg	0.94	0.94	0.94	422

Figure 16: Classification report for unoptimized model.

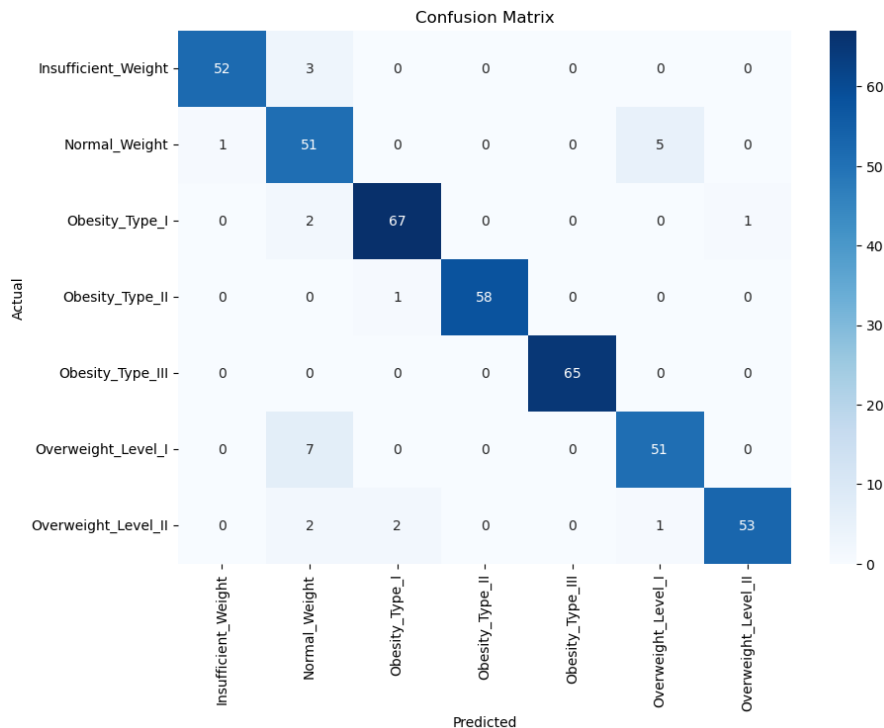


Figure 17: Confusion matrix for unoptimized model

From the above confusion matrix, it is evident that only major misclassification is between normal weight and overweight level 1 as 5 normal weight classes are misclassified as overweight level 1 and 7 overweight level 1 classes are misclassified as normal weight which is understandable as both are closely related.

6.4 CROSSVALIDATION

To better understand if the model actually performs better on unseen data, I performed crossvalidation using 5 folds which also resulted in the mean accuracy of 94%.

```
from sklearn.model_selection import cross_val_score

# Cross validation (5 folds)
cv_scores = cross_val_score(rf, x_train, y_train, cv=5, scoring='accuracy')

print(f"Cross-Validation Scores: {cv_scores}")
print(f"Mean CV Accuracy: {cv_scores.mean():.4f}")
print(f"Standard Deviation: {cv_scores.std():.4f}")

Cross-Validation Scores: [0.95266272 0.95857988 0.93491124 0.93768546 0.91988131]
Mean CV Accuracy: 0.9407
Standard Deviation: 0.0137
```

Figure 18: crossvalidation

6.5 HYPERPARAMETER TUNING

To improve the performance of the Random Forest model, I performed hyperparameter tuning using RandomizedSearchCV. A set of hyperparameters, such as the number of estimators, maximum depth, minimum samples split, minimum samples leaf, maximum features, and bootstrap method were randomly searched across 20 different combinations with 5-fold cross-validation.

After the search, the best parameters found were: 300 estimators, maximum depth of 30, minimum sample split of 2, minimum sample leaf of 1, max features as 'log2', and bootstrap set to False. These parameters were then used to retrain the final model for prediction which achieved an accuracy of 95% and the average precision, recall and f1 score was also 95%.

Classification Report:				
	precision	recall	f1-score	support
Insufficient_Weight	1.00	0.95	0.97	55
Normal_Weight	0.80	0.91	0.85	57
Obesity_Type_I	0.97	0.96	0.96	70
Obesity_Type_II	1.00	0.98	0.99	59
Obesity_Type_III	1.00	1.00	1.00	65
Overweight_Level_I	0.89	0.88	0.89	58
Overweight_Level_II	0.98	0.95	0.96	58
accuracy			0.95	422
macro avg	0.95	0.95	0.95	422
weighted avg	0.95	0.95	0.95	422

Figure 19: Classification report for optimized model.

7 STREAMLIT APP

7.1 HOME-PREDICTIONS

Users can input their personal details such as age, gender, height, weight, dietary habits, physical activity levels, and transportation modes through an intuitive form. After submitting the form, the inputs are preprocessed (scaling, encoding, transformation) and fed into the trained Random Forest model to predict the user's obesity level.

The screenshot shows the 'ObesitySense' Streamlit application interface. On the left is a sidebar with the app name and navigation options: 'Home - Prediction' (active), 'Dashboard', and 'Model Information'. The main area is titled 'ObesitySense: Obesity level Analysis and Prediction' and contains a form to 'Enter Your Details'. The form has two columns of inputs: Gender (Male), Age (23), Height (1.70), Weight (70), Family History (no), Eat high-calorie food? (no), Vegetable Consumption (Never), and Number of Main Meals (Between 1 & 2). The second column includes Eat Between Meals? (no), Do You Smoke? (no), Daily Water Intake (less than 1L), Monitor Calories? (no), Physical Activity (Never), Time Using Technology (0 to 2h), Alcohol Consumption (no), and Transportation (Automobile). A 'Predict' button is located at the bottom left of the form.

Figure 20: Streamlit app homepage.

7.2 DASHBOARD

This page provides interactive visualizations for deeper insights into the dataset. Users can filter the data based on gender and age range and explore distributions of obesity levels, important features like weight, vegetable consumption, physical activity, and water intake. It also includes a scatter plot of weight vs. height colored by obesity levels to reveal patterns visually.

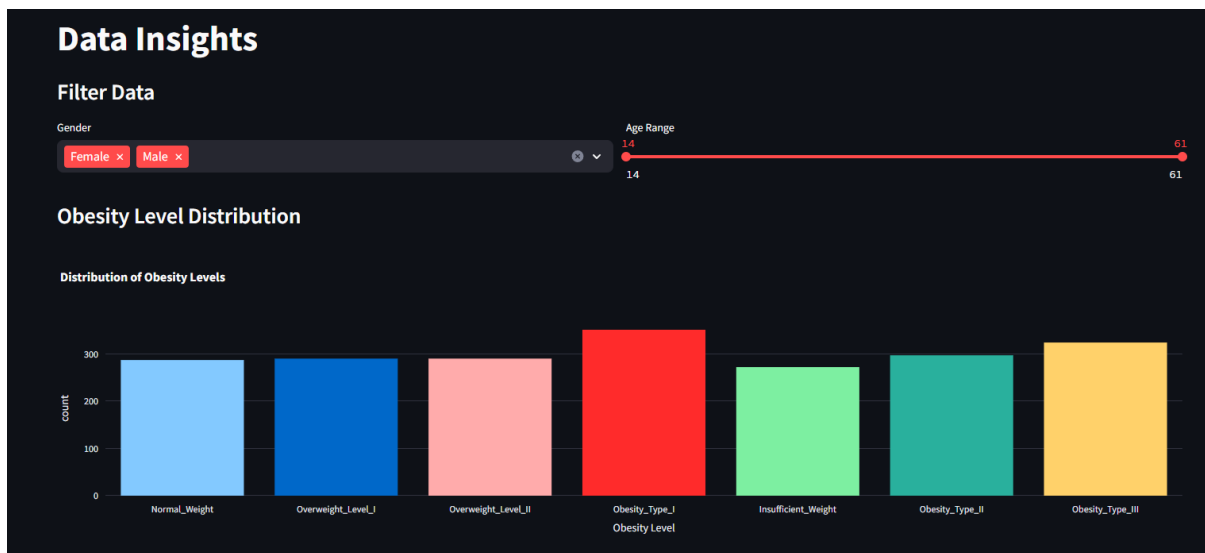


Figure 21: Dashboard page, Class distribution.

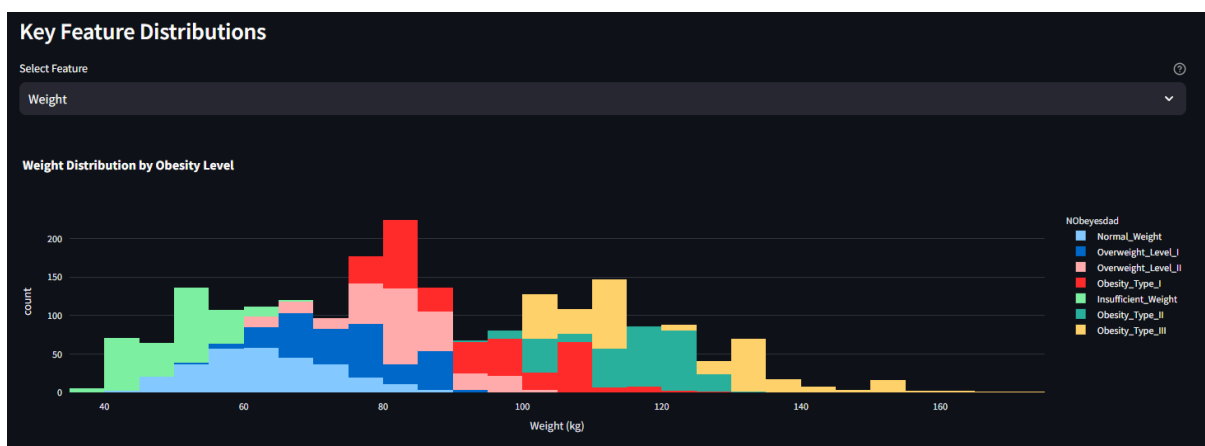


Figure 22: Dashboard page, weight distribution.

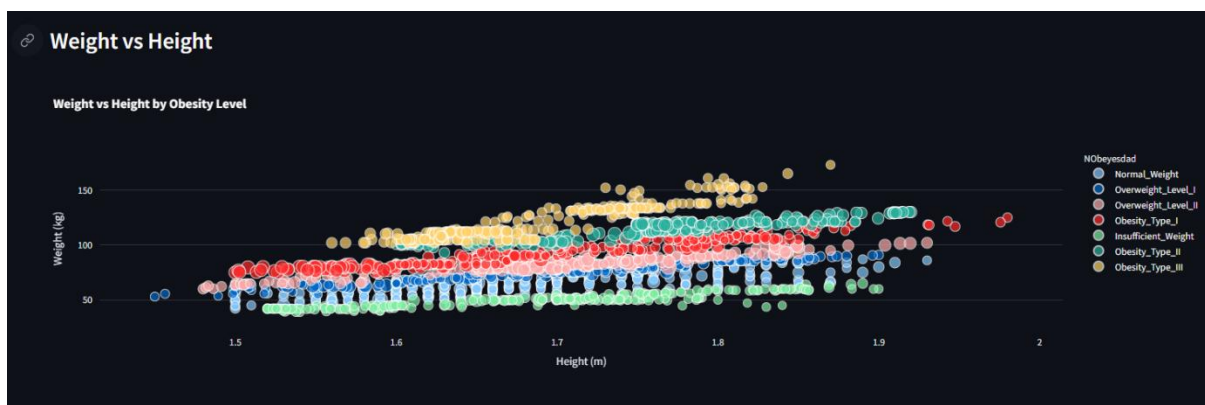


Figure 23: Dashboard page, Height vs Weight Scatterplot.

7.3 MODEL INFORMATION

This section explains the entire machine learning pipeline in a simple way. It covers:

- Data preprocessing techniques (like scaling, encoding methods),
- Model training and optimization steps.

- Model performance metrics (confusion matrix and scores).
- Feature importance, showing which factors most influenced the prediction results.

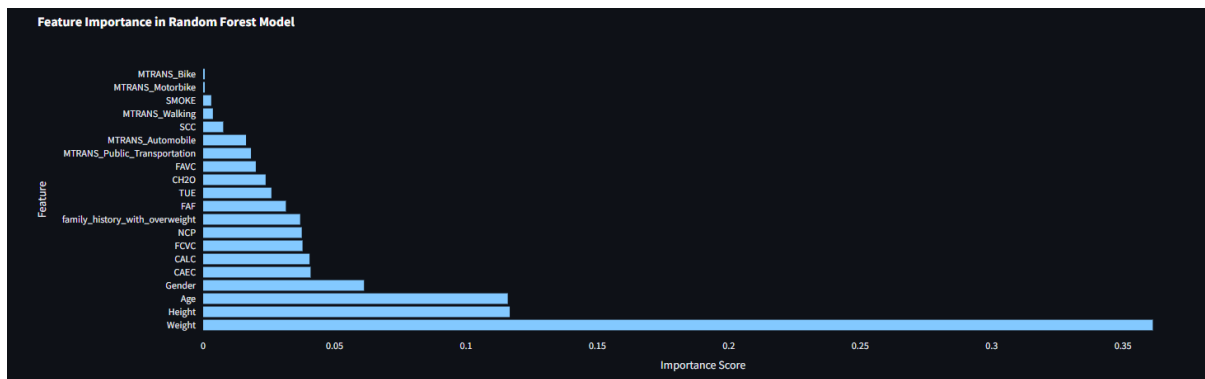


Figure 24: Feature importance extracted from RF model.

Overall, the app ensures that both technical and non-technical users can easily input their data, view predictions, understand the model, and explore the dataset.

8 EVALUATION

To assess the effectiveness of our obesity prediction model, we compared it with recent studies employing machine learning techniques. Our Random Forest model achieved an accuracy of 95%.

A study by Alghamdi et al. (2024) evaluated various machine learning models for obesity classification. Their Random Forest model, incorporating height and weight variables, achieved an area under the ROC curve (AUC) of 1.00, with a sensitivity of 0.96 and specificity of 0.99. Even without these variables, the model maintained a high AUC of 0.98, demonstrating the robustness of Random Forest in obesity prediction.

Similarly, Du et al. (2024) developed a visual obesity risk prediction system using the XGBoost algorithm. Their model effectively classified individuals into different obesity categories based on various health indicators, showcasing the applicability of machine learning in personalized health management.

In another study, Singh and Tawfik (2024) utilized data from the UK's Millennium Cohort Study to predict adolescent obesity. Employing machine learning models, they achieved a prediction accuracy exceeding 90%, highlighting the potential of such techniques in early obesity detection.

These studies underscore the efficacy of machine learning models, particularly Random Forest and XGBoost, in accurately predicting obesity. Our model's performance is consistent with these findings, reinforcing the reliability of our approach.

9 LIMITATIONS AND CHALLENGES

While the model performed well, there were some important limitations and challenges. First, the dataset used was partly synthetic, meaning it was artificially generated and may not fully

reflect real-world human behaviour and diversity. This could limit how well the model works with real patient data. Second, there was a gender-based class imbalance as some obesity categories were more common in one gender than the other, which might have caused biased predictions. Third, the model showed high dependence on the “Weight” feature, meaning it relied heavily on a single variable while possibly underestimating the importance of other health indicators. This could reduce the model’s accuracy for people with uncommon body types.

10 CONCLUSION

To conclude, the obesity level prediction model successfully demonstrated strong performance and provided valuable insights into the key factors behind obesity. While the model achieved high accuracy, the results highlighted areas for further improvement, particularly regarding data quality and feature dependency. Going forward, I plan to retrain the model excluding the Weight feature to test whether similar accuracy can be maintained using only lifestyle-related inputs. This will help assess the model's generalizability and potential for real-world deployment where some health metrics may be unavailable.

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APPENDIX I

To view the demo video, [CLICK HERE](#)

To visit the GitHub repository of the project, [CLICK HERE](#)